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Kelas : IF – 1

Rangkuman praktikum system terdesentralisasi dan terdistribusi

KOMUNIKASI ANTAR-SERVER (MESSAGE BROKER)

EC2 > Instans > Luncurkan instans

Luncurkan instans [Info](#)

Amazon EC2 memungkinkan Anda membuat mesin virtual, atau instans, yang berjalan di AWS Cloud. Segera mulai dengan mengikuti langkah-langkah sederhana di bawah ini.

Nama dan tanda [Info](#)

Nama

VM-1 Server

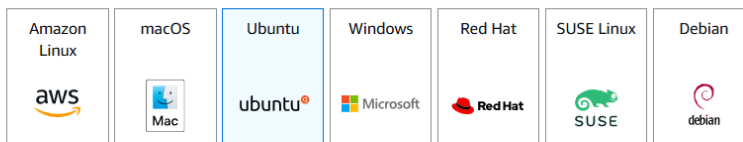
[Tambahkan tanda tambahan](#)

▼ Gambar Aplikasi dan OS (Amazon Machine Image) [Info](#)

AMI berisi sistem operasi, server aplikasi, dan aplikasi untuk instans Anda. Jika Anda tidak melihat AMI yang sesuai di bawah ini, gunakan bidang pencarian atau pilih [Telusuri AMI lainnya](#).

Terbaru

[Mulai Cepat](#)



[Jelajahi AMI lainnya](#)

Termasuk AMI dari AWS, Marketplace, dan Komunitas

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

Tingkat gratis yang memenuhi syarat

sgr-06d508010f60a420a

SSH	TCP	22	Kustom	0.0.0.0/0 X	Hapus
TCP Kustom	TCP	15672	IPv4-Di ...	0.0.0.0/0 X	Hapus
TCP Kustom	TCP	5672	IPv4-Di ...	0.0.0.0/0 X	Hapus

Tambahkan aturan

Aturan dengan sumber 0.0.0.0/0 atau ::/0 memungkinkan semua alamat IP mengakses instans Anda. Sebaiknya atur aturan grup keamanan untuk mengizinkan akses hanya dari alamat IP yang diketahui.

Batalan Pratinjau perubahan Simpan aturan

Pembahasan : membuat vm server rabbit dan client untuk grup keamanan nya dibuat SSH (port 22) TCP (port 15672) TCP (Port 5672) source anywhere

```
aws Cari [Alt+S] ID Akun: 1386-5493-4658 voclabs/user4457691=irvan.cahya24@students.utdi..
```

```
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

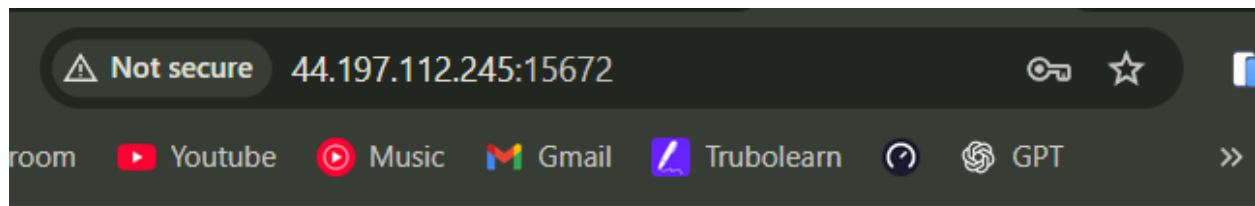
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-67-195:~$ sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1334 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1619 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [303 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.7 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1500 kB]
]
```

i-0b8c18d8d679c4710 (VM-1 Server)

PublicIPs: 44.197.112.245 PrivateIPs: 172.31.67.195





Username: *

Password: *

Pembahasan : instalasi server broker.

Jalankan perintah berikut untuk menginstal server pesan:

```
sudo apt-get update
```

```
sudo apt-get install -y rabbitmq-server
```

```
sudo rabbitmq-plugins enable rabbitmq_management
```

```
sudo rabbitmqctl add_user admin admin123
```

```
sudo rabbitmqctl set_user_tags admin administrator
```

```
sudo rabbitmqctl set_permissions -p / admin ".*" ".*" ".*"
```

Buka browser di laptop Anda, akses: <http://<PUBLIC-IP-VM-1>:15672> Login dengan user: admin dan

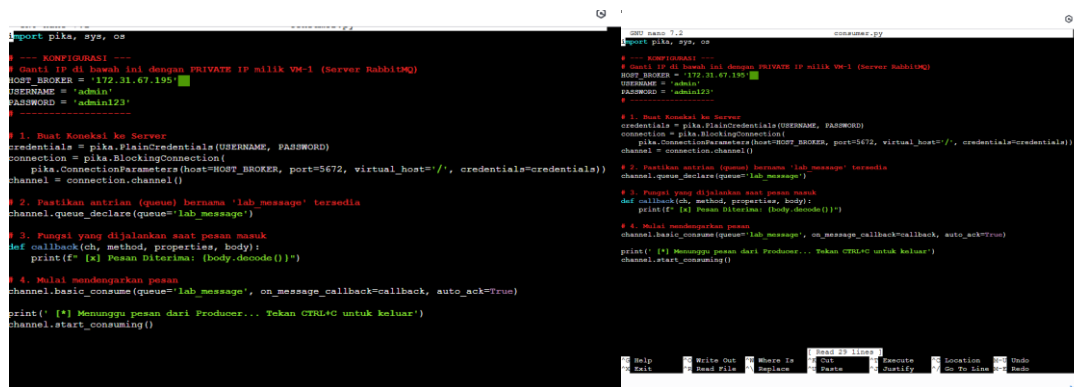
```

Get:22 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1334 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [218 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9444 B]
Get:26 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [909 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [205 kB]
Get:28 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:29 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.4 kB]
Get:30 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2194 kB]
Get:31 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [501 kB]
Get:32 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:33 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [5956 B]
Get:34 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Fetched 13.0 MB in 2s (5363 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-24-75:~$ sudo apt-get install -y python3-pika

```

i-000a8ff1c561f54d3 (Vm-client)

PublicIPs: 44.210.41.98 PrivateIPs: 172.31.24.75



```

# producer.py
import pika, sys, os

# --- KONFIGURASI ---
# Ganti IP di bawah ini dengan PRIVATE IP milik VM-1 (Server RabbitMQ)
HOST_BROKER = '172.31.67.195'
USERNAME = 'admin'
PASSWORD = 'admin123'

# 1. Buat koneksi ke Server
credentials = pika.PlainCredentials(USERNAME, PASSWORD)
connection = pika.BlockingConnection(
    pika.ConnectionParameters(host=HOST_BROKER, port=5672, virtual_host='/', credentials=credentials))
channel = connection.channel()

# 2. Pastikan antrian (queue) bernama 'lab_message' terdapat
channel.queue_declare(queue='lab_message')

# 3. Fungsi yang dijalankan saat pesan masuk
def callback(ch, method, properties, body):
    print(f' [x] Pesan Diterima: {body.decode()}')

# 4. Mulai mendengarkan pesan
channel.basic_consume(queue='lab_message', on_message_callback=callback, auto_ack=True)

print(' [*] Menunggu pesan dari Producer... Tekan CTRL+C untuk keluar')
channel.start_consuming()

# consumer.py
import pika, sys, os

# --- KONFIGURASI ---
# Ganti IP di bawah ini dengan PRIVATE IP milik VM-1 (Server RabbitMQ)
HOST_BROKER = '172.31.67.195'
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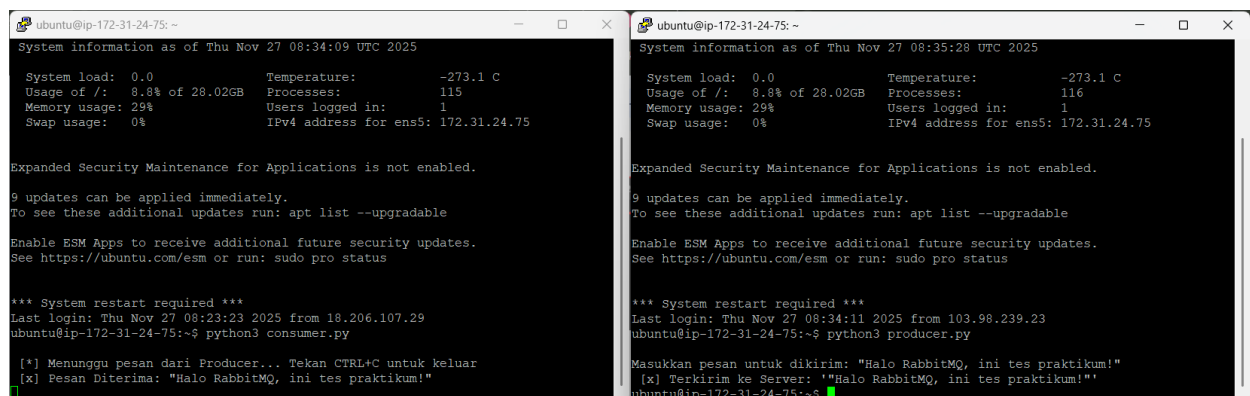
```

Pembahasan :di vm client sudo apt-get update

Menginstall library pika langsung dari repositori Ubuntu

sudo apt-get install -y python3-pika

setelah itu membuat script di vm2 consumer .py dan producer.py



```

ubuntu@ip-172-31-24-75:~$ sudo apt-get update
System information as of Thu Nov 27 08:34:09 UTC 2025

System load:  0.0          Temperature:   -273.1 C
Usage of /:   8.8% of 28.02GB Processes:      115
Memory usage: 29%         Users logged in: 1
Swap usage:   0%          IPv4 address for ens5: 172.31.24.75

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Thu Nov 27 08:23:23 2025 from 18.206.107.29
ubuntu@ip-172-31-24-75:~$ python3 consumer.py

[*] Menunggu pesan dari Producer... Tekan CTRL+C untuk keluar
[x] Pesan Diterima: "Halo RabbitMQ, ini tes praktikum!"

ubuntu@ip-172-31-24-75:~$ python3 producer.py
Masukkan pesan untuk dikirim: "Halo RabbitMQ, ini tes praktikum!"
[x] Terkirim ke Server: "Halo RabbitMQ, ini tes praktikum!"

```



Pembahasan : dengan menggunakan perintah python3 consumer.py python3 producer.py

Akan berjalan dan jika dilihat di rabbit akan terlihat lonjakan traffic menandakan praktikum berhasil

Kesimpulan : menggunakan load balancing pesan bisa didistribusikan dengan baik diantriakan di server, menggunakan round robin beban jadi tidka menumpuk

